

CS & IT ENGINEERING



Data Structure & Programming

Trees and Graphs

Lecture No.- 02



By- Satya sir

Recap of Previous Lecture



- Tree DS ?
- Terminology
 - Root, Leaf, Siblings
 - Degree, Level, Depth
 - Height
- Types of Binary Trees

Topics to be Covered



- Types of Binary Trees
- Tree Traversal
 - In order
 - Pre order
 - Post order





Topic : Tree Traversals

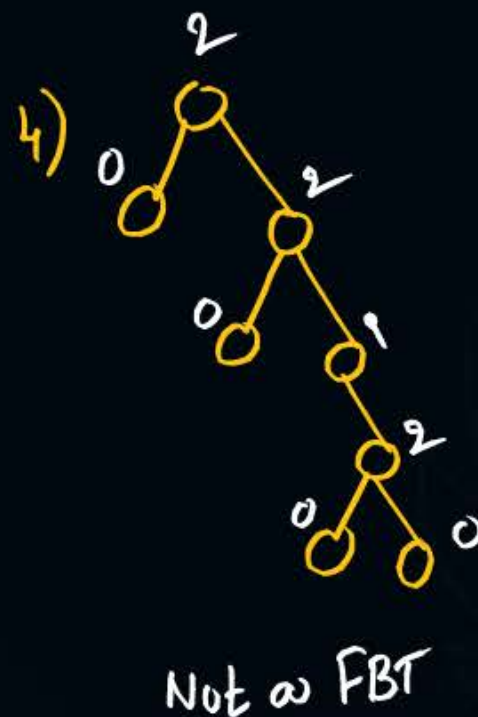
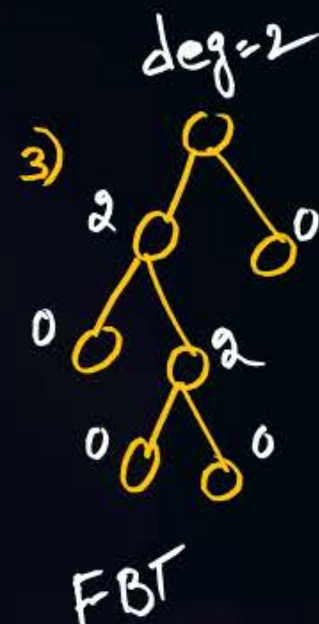
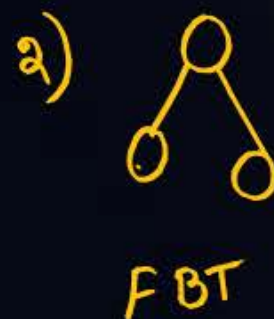
TG: SanyasirPW



Types of Binary Trees : FBT, CBT, PBT, SBT, DBT, BST, Heap, AVL Tree ...

Full Binary Tree : A Binary Tree in which, degree of Each node is Either 0 or 2.

Ex:





Topic : Tree Traversals



Complete Binary Tree (CBT)

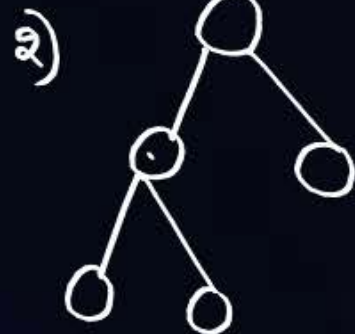
- A Binary Tree in which, Nodes are filled from top to bottom and Left to Right ($\downarrow \rightarrow$)

(OR)
- A Binary Tree in which all levels are completely filled, except last level.

Ex:
1)



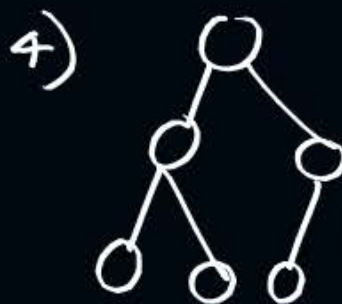
FBT, CBT



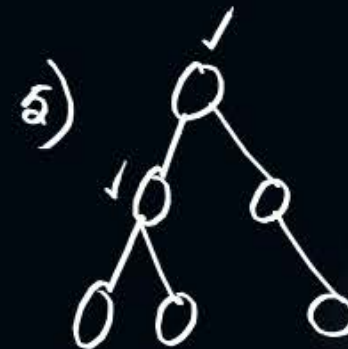
FBT, CBT



FBT, not a CBT



Not a FBT,
but, CBT



Not FBT
Not CBT



FBT, CBT

→ Every FBT need not to be CBT and Every CBT need not to be FBT.



Topic : Tree Traversals



Perfect Binary Tree (PBT)

- A Full Binary Tree in which all leaf nodes are at same level, is called PBT.

Ex:

①



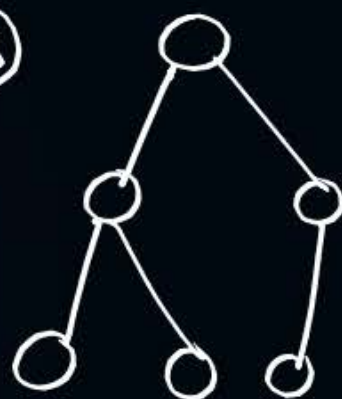
FBT, CBT,
PBT

2)



FBT, Not CBT
Not PBT

3)



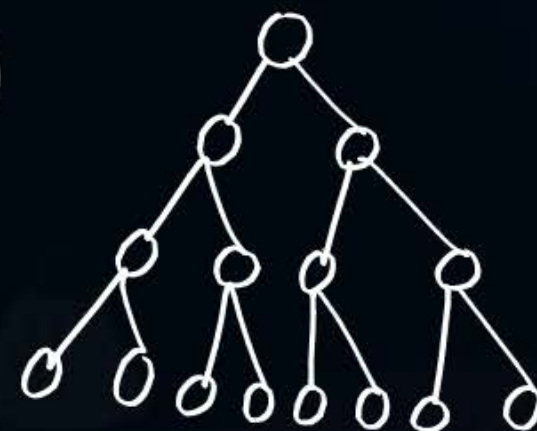
CBT, Not FBT
Not PBT

4)



CBT, FBT,
Not PBT

5)



CBT, FBT, PBT

→ Every FBT is not PBT, but Every PBT is FBT.

→ Every CBT is not PBT, but Every PBT is CBT.



Topic : Tree Traversals



Skewed Binary Tree (SBT)

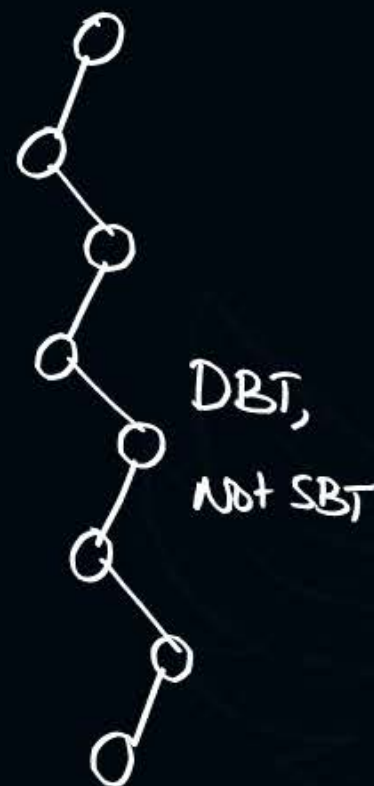
A Binary in which all nodes degree must be 1
Except leaf node, The child of all nodes must be
Only left child (or) only right child.



- Every SBT is DBT but Not Vice-Versa.

Degenerated Binary Tree (DBT)

A Binary Tree in which, all nodes degree must be 1, ^{Except leaf node.}
That one child can be Either left or Right.





Tree Traversals

⇒ The Sequence (order) of Visiting Nodes in a Binary Tree, such that every node is visited only one time.

⇒ 2 Types of Traversals

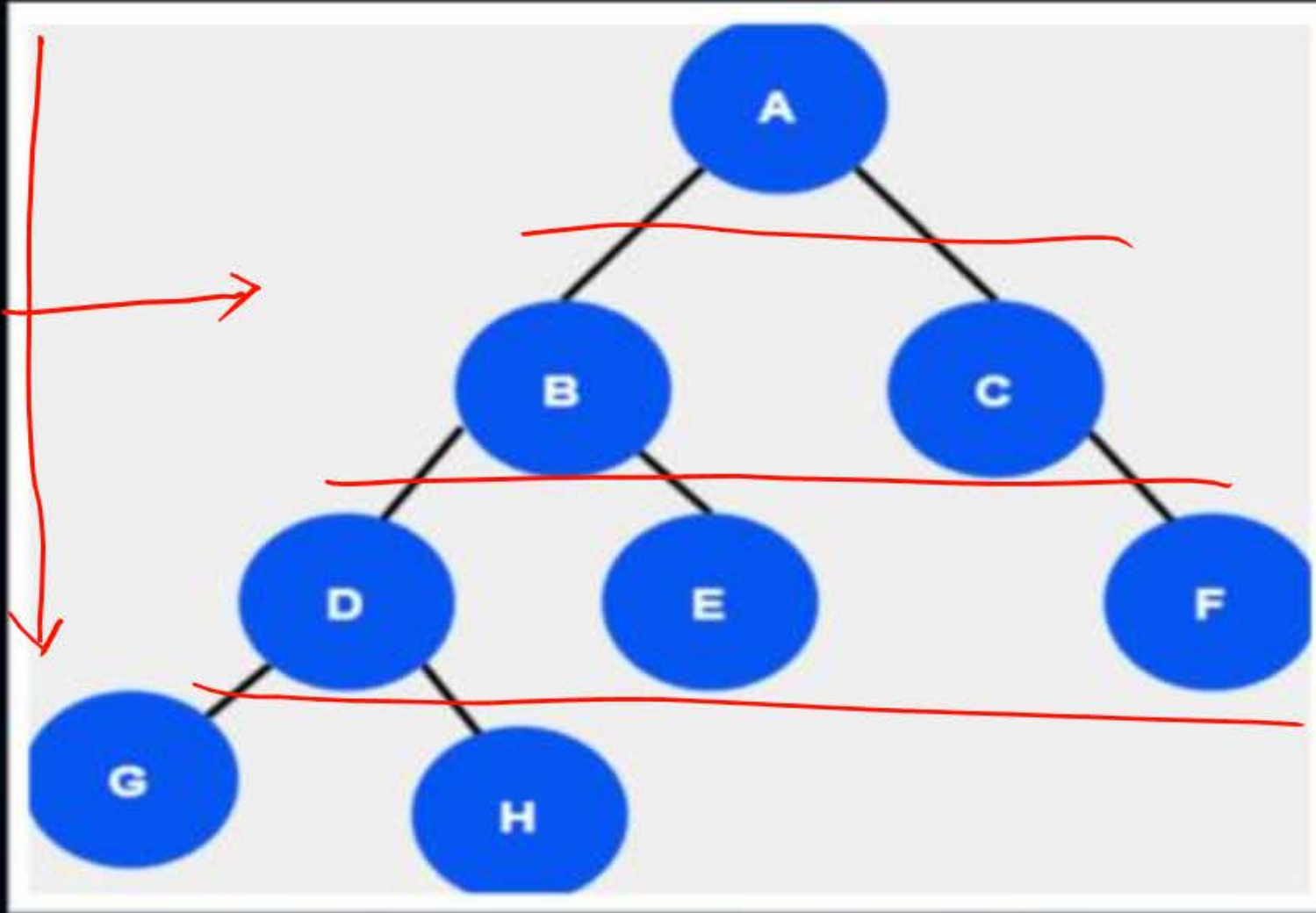
- Breadth-First Traversal : Level-order Traversal

- Depth-First Traversal

- In-order Traversal
 - Pre-order Traversal
 - Post-order Traversal



Topic : Tree Traversals



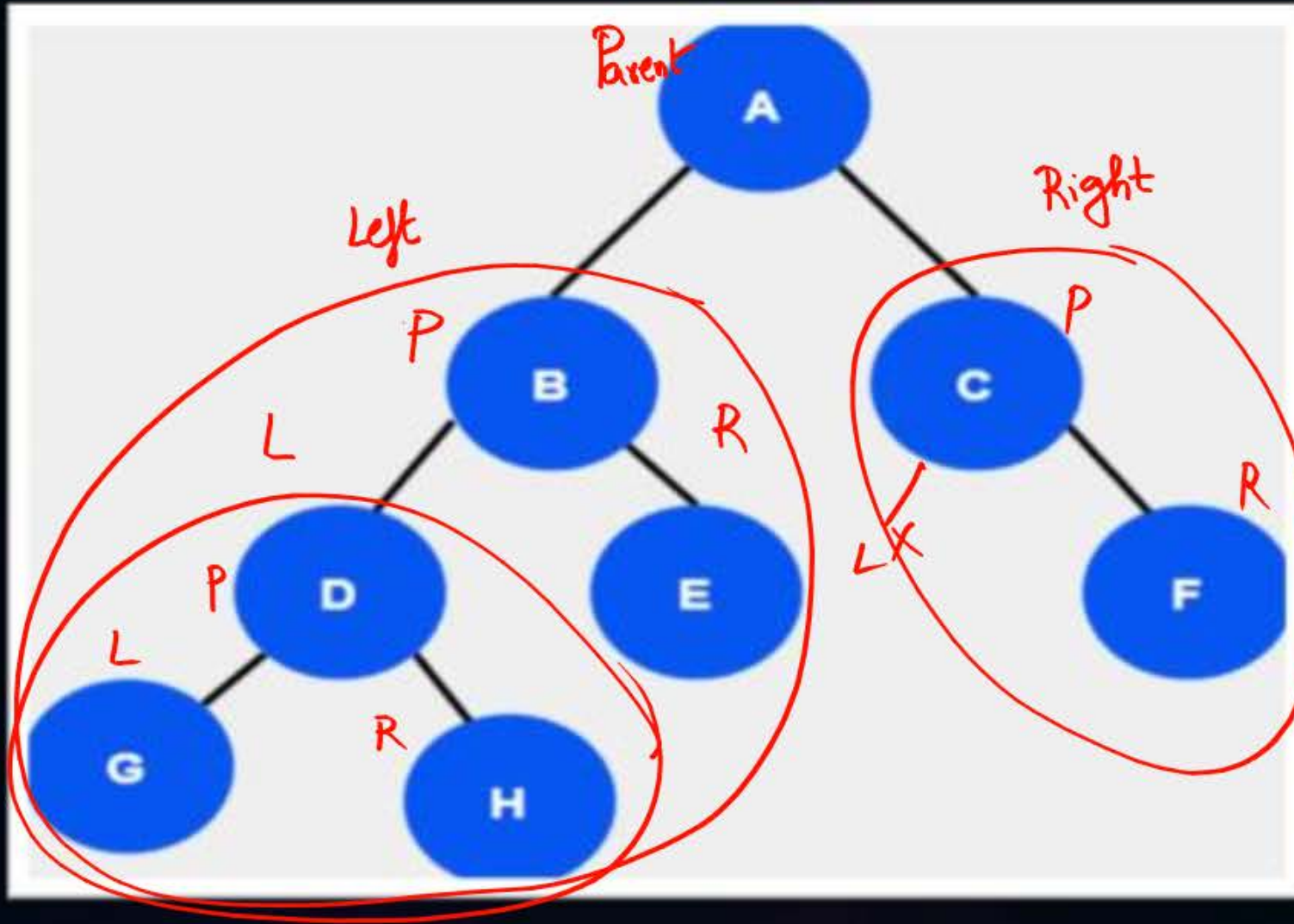
Breadth-First Traversal

- Level wise (top to bottom)
- Left to Right

A, B, C, D, E, F, G, H



Topic : Tree Traversals

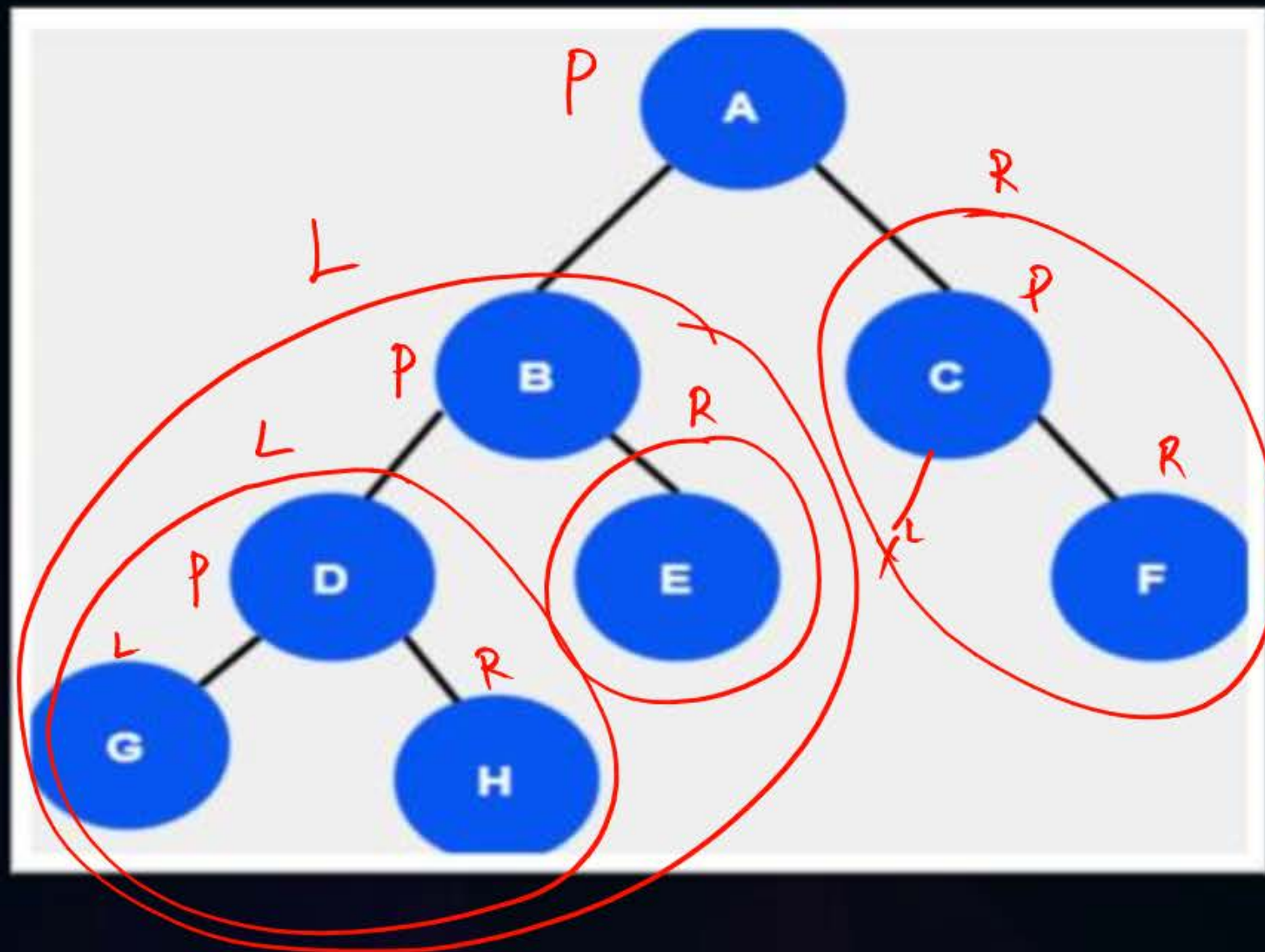


In order Traversal (LPR)
[Left child/subtree Parent Right child/subtree]

G, D, H, B, E, A, C, F



Topic : Tree Traversals



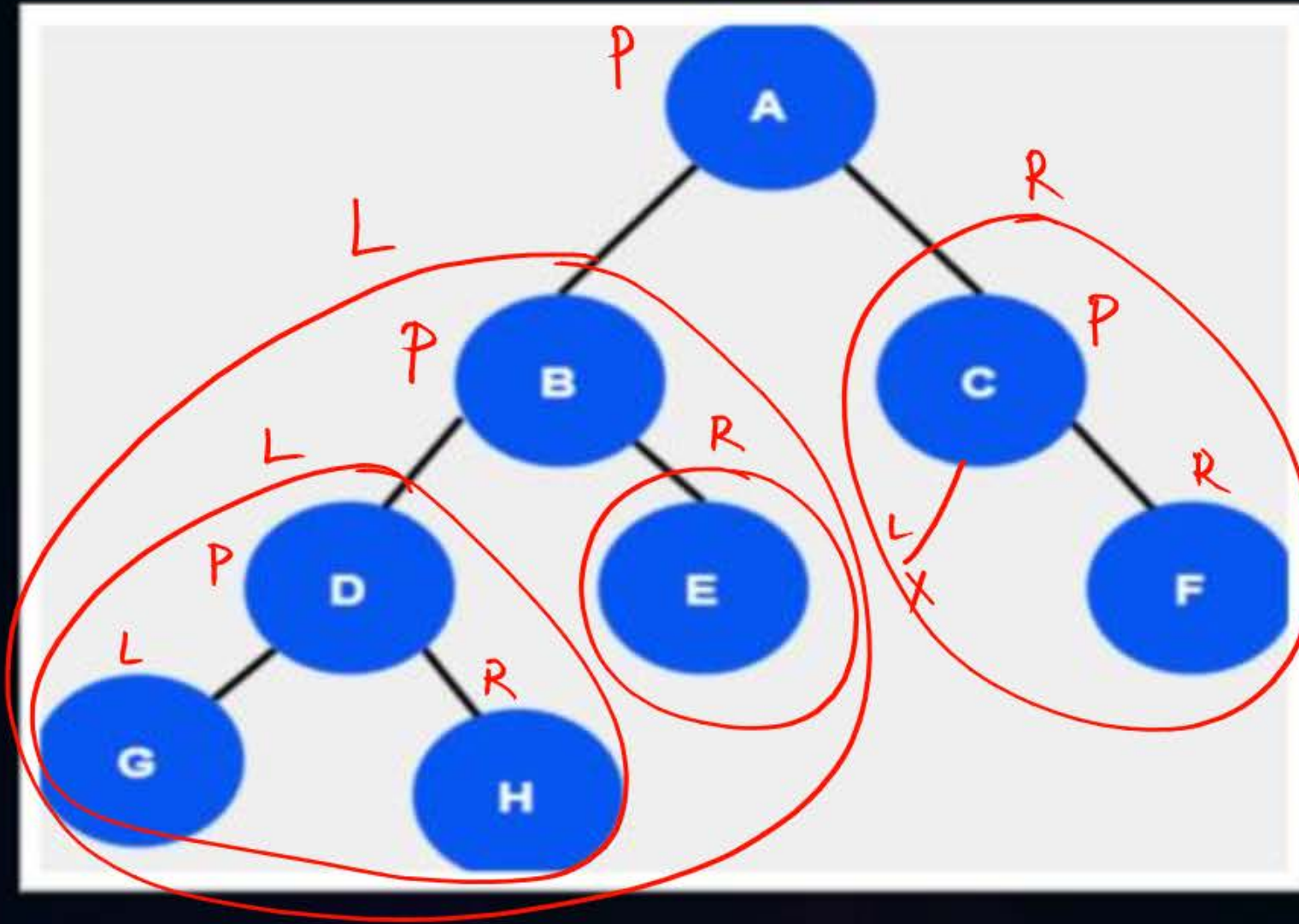
Pre-order Traversal [PLR]

[Parent Left Right]

A, B, D, G, H, E, C, F



Topic : Tree Traversals



Post-order Traversal [LRP]
[Left Right Parent]

G, H, D, E, B, F, C, A



Topic : Tree Traversals

Ex: 1



Level order

A B H C G I J D F K

E L M P N Q

In order (LPR)

D E C F B G A I

H Q P L K M N J

Pre order (PLR)

A B C D E F G H I J

K L P Q M N

Post-order (LRP)

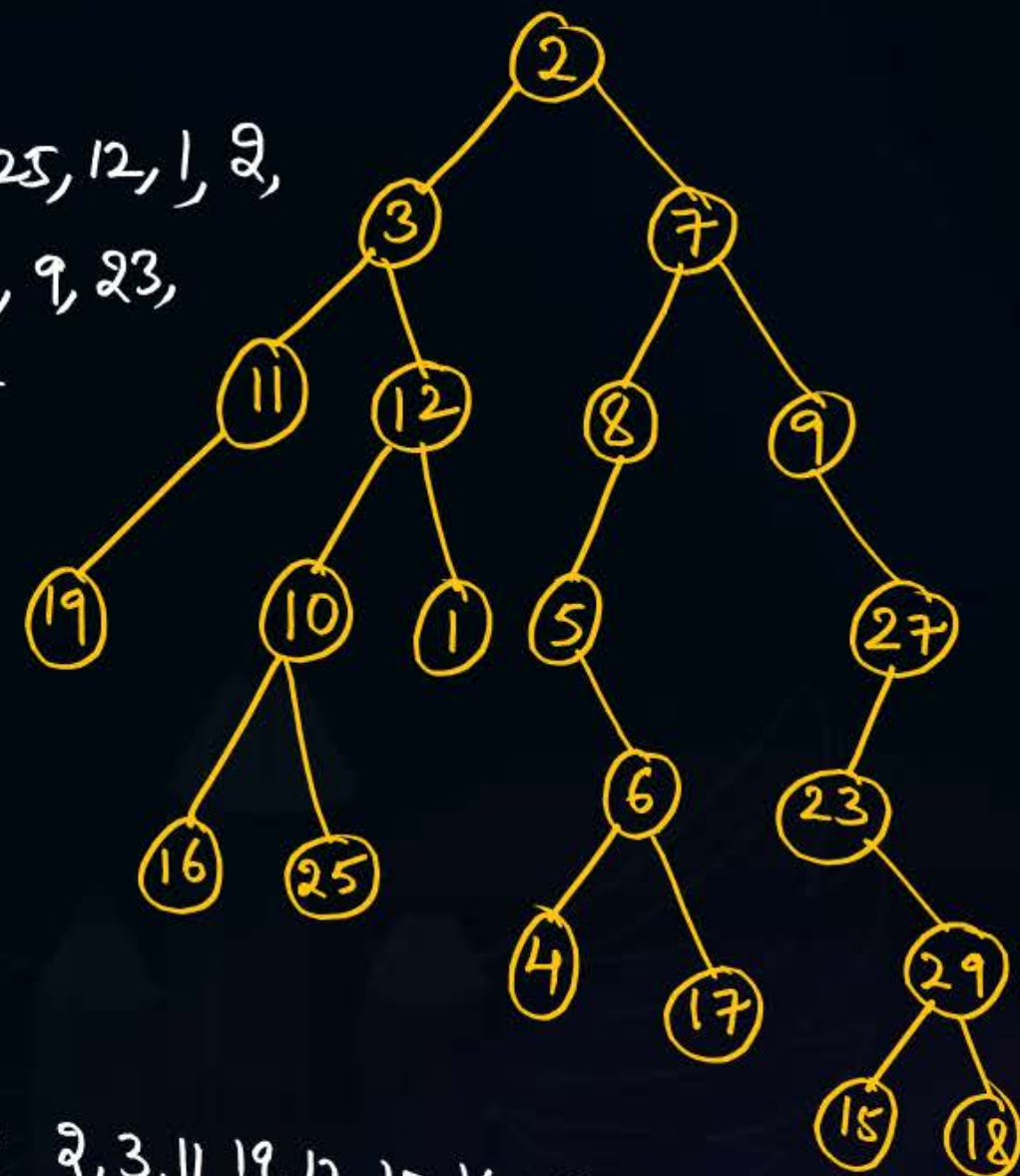
E D F C G B I Q P L

N M K J H A

Level: 2, 3, 7, 11, 12, 8, 9, 19, 10, 1, 5, 27,
16, 25, 6, 23, 4, 17, 29, 15, 18

In:

19, 11, 3, 16, 10, 25, 12, 1, 2,
5, 4, 6, 17, 8, 7, 9, 23,
15, 29, 18, 27



Pre order: 2, 3, 11, 19, 12, 10, 16, 25, 1, 7, 8,
5, 6, 4, 17, 9, 27, 23, 29, 15, 18

Post-order: 19, 11, 16, 25, 10, 1, 12, 3, 4, 17, 6, 5, 8,
15, 18, 29, 23, 27, 9, 7, 2





Topic : Tree Traversals



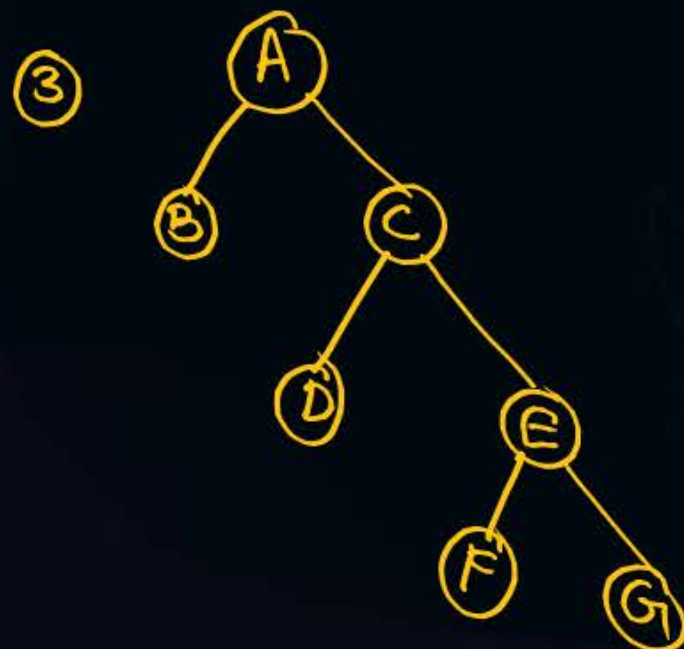
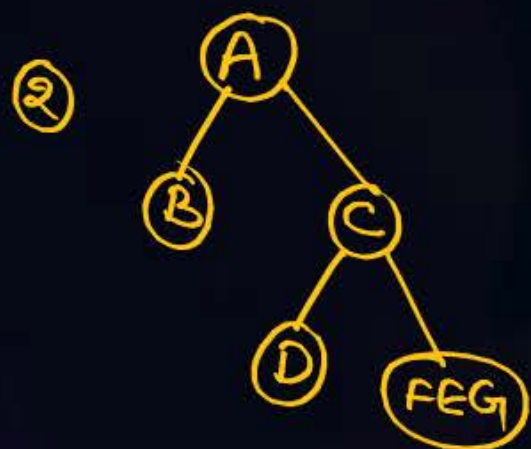
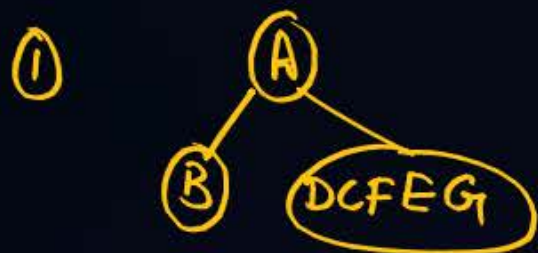
#Q. Consider In order Traversal: $\overset{L}{\boxed{B}} \cancel{A}, \overset{R}{\boxed{D, \cancel{C}, \cancel{F}, \cancel{E}, G}}$

(LRP) Post order Traversal: $\cancel{B}, D, \underline{F, G}, \cancel{E}, \cancel{C}, \underline{A}$

Pre order Traversal = _____

In order : To identify Left and Right children

Pre/Post order : To identify Parent Node.

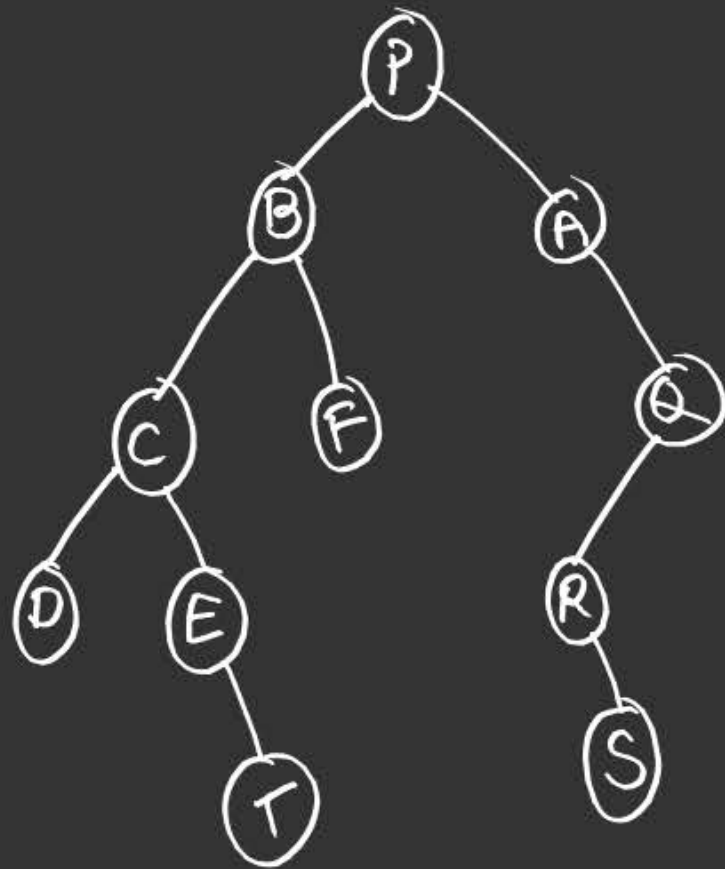


(PLR)
Pre order: A, B, C, D, E, F, G

#Q. In order: D, C, E, T, B, F, P, A, R, S, Q

(PLR) Pre order: P, B, C, D, E, T, F, A, Q, R, S

Post order: D, T, E, C, F, B, S, R, Q, A, P





2 mins Summary



- Types of Binary Trees
 - FBT, CBT, PBT, SBT, DBT
- Tree Traversals
 - BFT, DFT (In, Pre, Post)

To be contd...





THANK - YOU